DAY 3: CONFIGURING RIP

RIP Concepts, Multi-Router Configuration & Simulation Analysis

Theory: RIP Concepts

RIP (Routing Information Protocol): RIP is a distance-vector dynamic routing protocol. Routers periodically exchange routing information and select routes mainly by hop count.

Hop Count: Hop count is the number of routers a packet must pass through to reach a destination. RIP supports a maximum usable metric of 15 hops; a metric of 16 represents an unreachable network.

Updates: RIP routers normally send periodic routing updates approximately every 30 seconds. RIPv2 sends updates using multicast address 224.0.0.9 and supports classless routing information.

Convergence: Convergence is the process by which routers reach a consistent view of network reachability after a topology change. RIP can converge more slowly than modern link-state protocols.

Task 1: Three-Router Topology & IP Addressing

PC1 — SW1 — R1 ===== R2 ===== R3 — SW3 — PC3

Use serial links between routers (or Ethernet if preferred). The following addressing plan uses serial /30 point-to-point networks.

Device	Interface	IP Address	Subnet Mask
R1	G0/0	192.168.1.1	255.255.255.0
R1	S0/0/0	10.0.12.1	255.255.255.252
R2	S0/0/0	10.0.12.2	255.255.255.252
R2	S0/0/1	10.0.23.1	255.255.255.252
R3	S0/0/0	10.0.23.2	255.255.255.252
R3	G0/0	192.168.3.1	255.255.255.0
PC1	NIC	192.168.1.10	255.255.255.0
PC3	NIC	192.168.3.10	255.255.255.0

PC1 gateway: 192.168.1.1 | PC3 gateway: 192.168.3.1

R1 - Interface Configuration

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit
interface serial 0/0/0
 ip address 10.0.12.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit
end
```

R2 - Interface Configuration

```
enable
configure terminal
hostname R2
```

```
interface serial 0/0/0
 ip address 10.0.12.2 255.255.255.252
 no shutdown
 exit
interface serial 0/0/1
 ip address 10.0.23.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit
end
```

R3 - Interface Configuration

```
enable
configure terminal
hostname R3
interface serial 0/0/0
 ip address 10.0.23.2 255.255.255.252
 no shutdown
 exit
interface gigabitEthernet 0/0
 ip address 192.168.3.1 255.255.255.0
 no shutdown
 exit
end
```

Important: 'clock rate 64000' is required only on the DCE end. Interface names can differ by router model; verify them using 'show ip interface brief'.

Task 2: Configure RIP Version 2

R1:

```
configure terminal
router rip
 version 2
 no auto-summary
 network 192.168.1.0
 network 10.0.0.0
 exit
end
write memory
```

R2:

```
configure terminal
router rip
 version 2
 no auto-summary
 network 10.0.0.0
 exit
end
write memory
```

R3:

```
configure terminal
router rip
 version 2
 no auto-summary
 network 10.0.0.0
 network 192.168.3.0
 exit
end
write memory
```

Verification

```
show ip interface brief
show ip route
show ip route rip
show ip protocols
```

A RIP-learned entry may look like:

```
R 192.168.3.0/24 [120/2] via 10.0.12.2
```

R = RIP, 120 = RIP administrative distance, and 2 = hop-count metric.

Task 3: Observe a Hop-Count Change

Insert an additional router (R4) between R2 and R3. The original path R1 → R2 → R3 becomes R1 → R2 → R4 → R3. Re-address the affected point-to-point links and enable RIPv2 on R4.

Before: R1 → R2 → R3 | After: R1 → R2 → R4 → R3

Example new links:

```
R2-R4 network: 10.0.24.0/30
R2 S0/0/1:     10.0.24.1/30
R4 S0/0/0:     10.0.24.2/30

R4-R3 network:  10.0.43.0/30
R4 S0/0/1:     10.0.43.1/30
R3 S0/0/0:     10.0.43.2/30
```

R4 RIP configuration:

```
enable
configure terminal
hostname R4
router rip
 version 2
 no auto-summary
 network 10.0.0.0
 exit
end
```

After configuring the new interfaces and RIP, run on R1:

```
show ip route
show ip route rip
```

Observation	Before adding R4	After adding R4
Path to 192.168.3.0/24	R1 → R2 → R3	R1 → R2 → R4 → R3
Expected RIP metric from R1	2 hops	3 hops
Actual value observed	_____	_____

Task 4: Capture RIP Updates in Simulation Mode

1. Open Cisco Packet Tracer and switch from Realtime to Simulation mode.

2. Open Edit Filters / Event List Filters and select RIP (you may clear other protocols to reduce noise).

3. Use Capture/Forward or Auto Capture/Play and wait for RIP update events between routers.

4. Click a RIP event/envelope and inspect the PDU details.

5. Record the source router/interface, destination, and routes/metrics advertised in the update.

6. For RIPv2 on IPv4, observe that routing updates are normally sent to multicast address 224.0.0.9.

Packet	Source	Destination	Advertised network	Metric
1	_____	_____	_____	_____
2	_____	_____	_____	_____
3	_____	_____	_____	_____

Analysis / Conclusion

RIP automatically exchanges routing information between neighboring routers. The routing table uses hop count to select routes, and adding another router to the path increases the RIP metric. Packet Tracer Simulation mode makes the RIP update process visible and allows the source, destination, advertised networks, and metrics to be inspected. When the topology changes, RIP updates eventually allow routers to converge on the new network information.

Useful Commands

```
show ip interface brief
show ip route
show ip route rip
show ip protocols
ping <destination-IP>
show running-config | section router rip
```